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RADIATIVE LIFETIMES OF THE $B^1\Pi$ STATE OF LiH AND LiD

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The lifetimes of the $B^1\Pi$ vibrational levels of LiH and LiD were measured in a molecular beam apparatus using the method of delayed coincidences. The results for LiH are: $\tau(v'=0)=11.4$ ns, $\tau(v'=1)=17.0$ ns, $\tau(v'=2)=24$ ns and for LiD $\tau(v'=0)=11.0$ ns, $\tau(v'=1)=14.0$ ns, $\tau(v'=2)=21$ ns. These values are in good agreement with theoretical calculations. Three new bands of the A–X system were identified.

1. Introduction

The existence of four LiH isotopes with significantly different reduced mass allowed an accurate examination of fundamental concepts such as reduced quantum numbers, Born–Oppenheimer approximation, adiabatic and non-adiabatic effects.

In the past we measured lifetimes and dipole moments of the $A^1\Sigma^+$ state of LiH and LiD. A detailed analysis of the dipole moment of the $v'=5$ level of LiH led to the assumption that even non-adiabatic corrections have to be taken into account for a correct description of the measured splitting [1].

We now started to examine the B state of LiH and LiD. Velasco [2] first observed the absorption spectrum of the B–X transitions of these isotopes in the early fifties. Some additional data were measured by Stwalley et al. in 1971 [3]. Zemke et al. determined potential curves for the X, A and B states of LiH [4]. They joined the RKR data for low vibrational levels to the long-range region by scaled ab initio results of Docken and Hinze [5]. With this potential they calculated transition probabilities [6] and radiative lifetimes [7]. In 1982 Vidal and Stwalley published improved potential curves with adiabatic corrections for all isotopic species using the IPA method [8].

In this work, the radiative lifetimes of the B state were measured experimentally for the first time and theoretical lifetimes were calculated. We recorded the LiH spectrum between 289 and 305 nm. In addition to the (2, 0), (1, 0), (0, 0) and (2, 1), (1, 1) (0, 1) bands of the B–X transitions reported by Velasco, we identified the (23, 0), (22, 0), (21, 0) bands of the A–X transition.

2. Experimental

The lifetimes were measured by observation of delayed coincidences of molecular fluorescence in a molecular beam, i.e.

$$\mathrm{LiH}(X^1\Sigma^+) + h\nu_1 \rightarrow \mathrm{LiH}^*(B^1\Pi) \rightarrow \mathrm{LiH}(X^1\Sigma^+) + h\nu_2,$$

$$\mathrm{LiD}(X^1\Sigma^+) + h\nu_1 \rightarrow \mathrm{LiD}^*(B^1\Pi) \rightarrow \mathrm{LiD}(X^1\Sigma^+) + h\nu_2.$$

The production of LiH (and LiD) was realized in a high-grade steel oven. Metallic lithium was heated in the presence of an argon gas flow up to 800 K. Then, a H_2 gas flow was run through the lithium melt and the temperature was increased up to about 950 K. The molecules left the oven through a 1 mm nozzle, and formed an effusive molecular beam which passed through three apertures into the reaction chamber.

The output of an excimer laser pumped dye laser (EMG50E and FL2002E of Lambda Physik) was frequency doubled with a KDP crystal. This system produced UV laser pulses with 4 ns pulse

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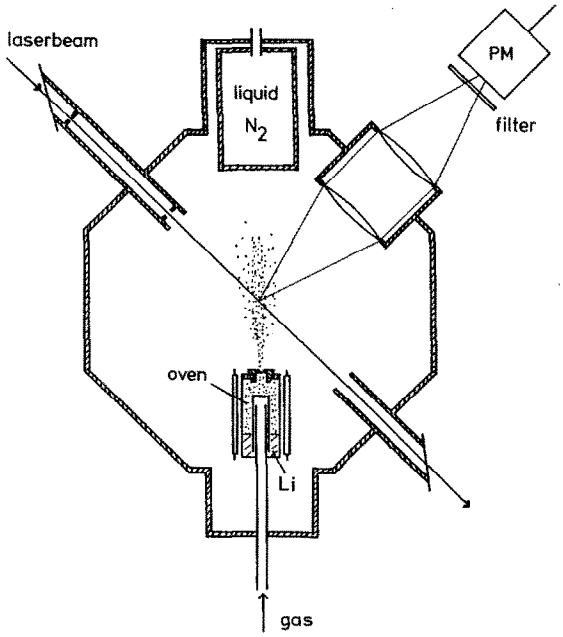


Fig. 1. Schematic diagram of the experimental apparatus.

width and a peak power of 40 kW. The maximum repetition rate was 200 Hz. Fig. 1 shows the geometry of the experiment.

The laser beam crossed the molecular beam at an angle of 135° and the fluorescence of the excited molecules was observed perpendicularly to the laser beam.

The scattered laser light and the radiation of the oven were cut off by two Schott filters BG 18 and UG 1. They allowed the fluorescence between 320 and 400 nm to pass, which was detected by a cooled RCA 31034 photomultiplier. The lifetimes were measured by the method of delayed coincidences. At a rate of about 0.1 photon per laser pulse, the time interval between laser shot and fluorescence photon is converted in an amplitude (Ortec 437A) which is registered in a multichannel analyser (Ortec 6240B). A typical curve is shown in fig. 2.

A good time calibration of the electronic system was achieved with a fast pulse generator. We confirmed this calibration by a measurement of the well-known lifetime of the Na $3p^2P_{1/2}$ state. After a pile up correction and a deconvolution with the excitation pulse, the decay curve was least-squares fitted to a single exponential decay. For the survey spectrum the LiH was produced in a cross cell. The fluorescence of a volume of about $5 \times 1.5 \times 1.5$ mm^3 was imaged to a Jobin Yvon H20hV monochromator (1 mm slits) followed by a 1P28A photomultiplier. We used the line positions given by Velasco [2] as frequency reference. The line intensities are not representative because the UV radiation power was not constant during the wavelength scan.

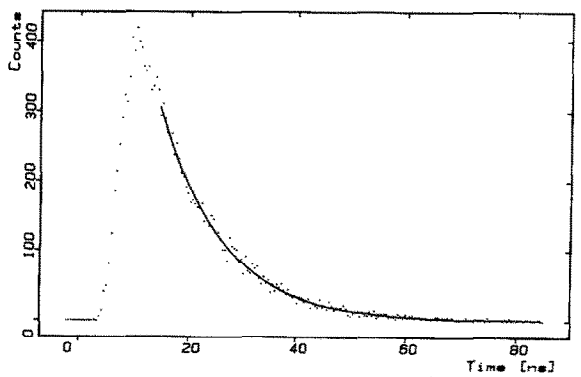


Fig. 2. Fluorescence decay curve of the LiH $B^1\Pi(v' = 0)$ state fitted with a single exponential decay. Excitation line: R(1) of the (0, 0) band of the B–X transition.

3. Results

Despite the small interaction volume (compared to a 16 m absorption cell) and the small transition probability, the (0, 0) and (1, 0) bands of LiH and LiD could be recorded with good S/N ratio. The intensity decreased towards the (2, 0) band and was clearly lower for LiH than for LiD. The band origin of the (1, 0) and (2, 0) bands of LiH are shown in fig. 3.

As mentioned above we identified the (23, 0) (22, 0) and (21, 0) bands of the A–X transition, as well as the origin of (20, 0), which was not detected in ref. [9]. Newly identified lines are given in table 1. The ΔG values obtained from these data and from ref. [9] are listed in table 2 and are compared with theoretical calculations of Partridge and Langhoff [10] and Stwalley et al. [4,11]. The results of the hybrid potential of Stwalley et al. [4]

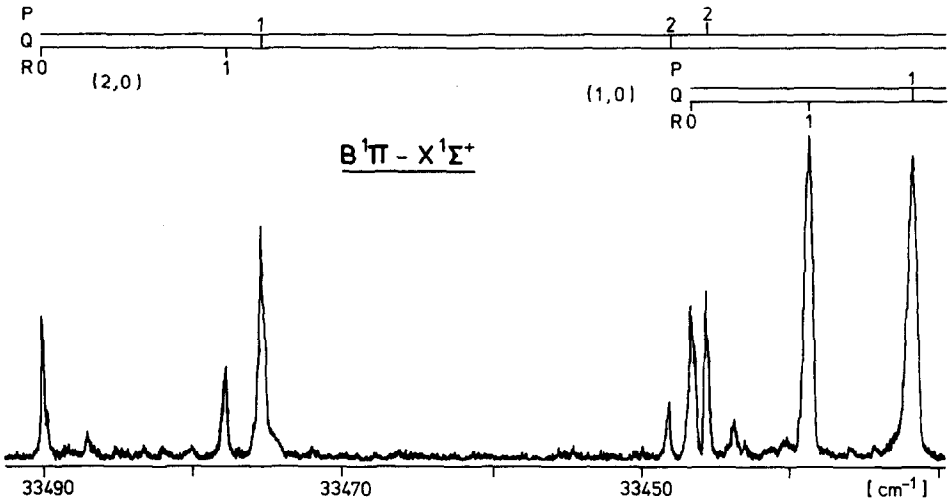


Fig. 3. Section of the LiH spectrum, showing the band origins of the B–X (1, 0) and (2, 0) bands.

show the best agreement with our measurements.

For the determination of the lifetimes we measured several R and Q lines. The results are figured in table 3. The determination of the $v' = 2$ state lifetimes is difficult because of the weak lines especially for LiH. In an experiment with much scattered laser light, we measured a lifetime of 25.8 ns for the R(0) and 21 ns for the Q(1) line of (2, 0) LiH, the latter being the strongest line of this band. The error of these values is at least 20% and additionally there may be an overlap between the Q(1) line and a Li_2 line. In a cell measurement at a pressure of 1 mbar the lifetime of the $v' = 2$ state using Q(1) excitation was much shorter than the lifetime with R(0) excitation. Lines with $v' = 3$ (which are possible for LiD) have not been observed. Zemke and Stwalley [7] calculated theoretical lifetimes for the B state of LiH. For a comparison with our results, we needed also the LiD lifetimes. For this purpose, we calculated the vibrational wavefunctions by integrating the one-dimensional Schrödinger equation using the Numerov–Cooley method with the hybrid potential curves of Stwalley et al. [4]. Using the transition moment functions of Docken and Hinze [5] (extended by Zemke et al. [6]), we obtained the

Table 1
Lines of the $A^1\Sigma^+$–$X^1\Sigma^+$ band system of ^{7}LiH (in cm^{-1})

J	(20, 0) band		(21, 0) band		(22, 0) band		(23, 0) band	
	R branch	P branch	R branch	P branch	R branch	P branch	R branch	P branch
0	33183.3		33487.7		33769.4			
1	33177.0	33164.3	33481.1	33468.8	33762.1	33750.9		
2	33160.5	33147.8	33463.5	33443.1	33743.6	33725.1	33994.5	
3	33133.5	33103.1	33435.3	33407.0	33714.1	33688.4		33940.0
4	33095.3	33057.1	33396.4	33360.1	33674.0	33640.2	33919.9	33891.0
5	33047.4	33000.2	33346.6	33302.7	33622.2	33581.5	33867.3	33830.8
6			33286.1	33234.3	33559.9	33512.1	33801.8	33759.2
7			33215.0	33156.0	33486.4	33431.6	33725.1	33676.5
8			33133.5	33066.9	33401.9	33340.5	33636.8	33582.6
9			33041.4		33306.4	33238.7	33537.0	33477.5
10					33199.5	33126.1	33425.6	33361.0
11					33082.4		33306.4	33233.8

Table 2
Comparison of calculated and observed vibrational level spacings $G_{v+1} - G_v$ in the $A^1\Sigma^+$ state of LiH in cm^{-1}

v	Observed	Calculated		
		ref. [10]	ref. [4]	ref. [11]
19	323.2±0.5	330.4	323.7	318.2
20	304.7±0.5	313.1	305.9	298.8
21	282.0±0.5	291.8	284.0	274.3
22	253.5±0.5	264.4	256.9	243.1

Table 3
Lifetimes of the vibrational states of LiH and LiD

v'	LiH			LiD		
	exp.	obs. lines	theor.	exp.	obs. lines	theor.
0	11.4±0.6	R(0),Q(5), R(1)	11.5	11.0±0.6	Q(5)	11.0
1	17.0±0.8	R(1)	17.2	14.0±0.8	R(3), Q(2)	14.7
2	23.4±4.0 [a]	R(0),Q(1)	24.0	21.0±1.2	Q(3)	19.8
3						24.8

[a] The large error of this value is explained in the text.

transition probabilities for the B–X and B–A transitions. The resulting lifetimes are essentially caused by the B–X transition. Our calculated LiH lifetimes agree within 2% with the values given in ref. [7]. The differences may be caused by interpolating the potentials using a tensioned cubic spline fit with different tension parameters at critical points. The existence of the highest vibrational level and therefore the bound–free contribution to the lifetimes depend on the choice of this parameter.

Nevertheless, this shows only a small effect on the lifetimes due to the scaling of the bound–free contribution [7].

4. Discussion

The experimental values are in good agreement with theoretical calculations. Our results confirm the potential curves and the ab initio calculation of Docken and Hinze [5]. The experimental error

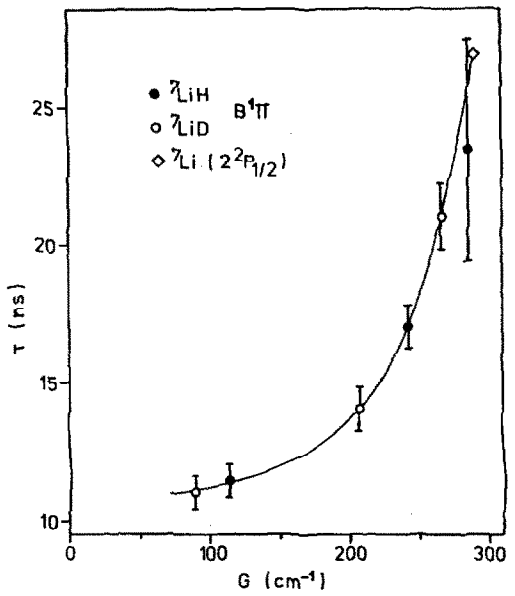


Fig. 4. LiH and LiD lifetimes as a function of the energy of the corresponding vibrational levels. The solid line is only a guide for the reader's eye.

for $v' = 2$ is too large to determine the bound–free contribution exactly. The lifetime is increasing with v'. It shows a tendency towards the lifetime of the $2\,^2P_{1/2}$ state of Li (26.9 ns). The LiD lifetimes show the same tendency but they are shorter than for LiH. This corresponds to the stronger lines in the spectrum. It is instructive to plot the lifetime of the vibrational levels as a function of their energy in the $B^1\Pi$ state. Within the Born–Oppenheimer approximation these lifetimes are mainly determined by the position of the corresponding level in the $B^1\Pi$ potential. The influence of the lower electronic states should be small because of the large number of vibrational levels in the deep A and X potentials. Hence the lifetimes should show a regular dependence on the vibrational energy. This is illustrated in fig. 4 for the two isotopes LiH and LiD.

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